Decision Assistance and Steering in Health Insurance

Economics brown bag

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Motivation

Intermediaries and potential conflicts of interest



88% of home buyers purchased through an agent or broker

NAR paid $418 million to settle home sellers' claims over broker commissions. Offers of buyer-broker compensation can no longer be posted on the MLS.

NAR 2025 Profile of Home Buyers and Sellers; NAR settlement, approved Nov. 2024

Intermediaries and potential conflicts of interest



Housing

realtors



60% of non-retirement investors use financial advisers

7% of financial advisers have misconduct records ($>$ 15% at some of the largest firms), with relatively minor labor market consequences.

Egan, Matvos, and Seru, *JPE* 2019; *JEP* 2024

Intermediaries and potential conflicts of interest



Housing

realtors



Investing

financial advisers



43.7% of new Medicare Advantage enrollees used an agent in 2022

$10.1 billion in agent commissions that year. In the ACA exchanges, 78% of active enrollments were agent-facilitated in 2024.

Meyers et al., *JAMA Internal Medicine* 2026; Gürel, *Risk Management and Insurance Review* 2024

Intermediaries and potential conflicts of interest



Housing

realtors



Investing

financial advisers



Insurance

agents

help consumers choose a product

or

steer them toward products that pay more

Intermediaries and potential conflicts of interest



Housing

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Insurance

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Our Research Question: What are the welfare effects of assistance and steering in health insurance?

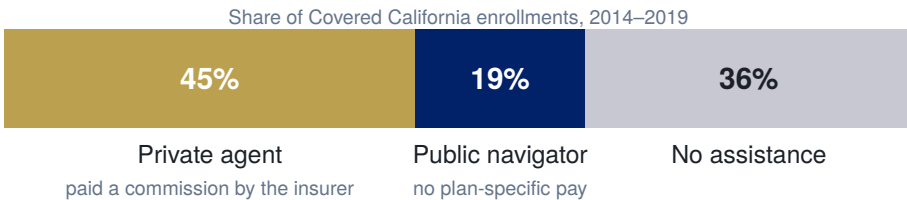
Our Setting: Covered California ACA Exchanges

Share of Covered California enrollments, 2014–2019



Private agent	Public navigator	No assistance
paid a commission by the insurer	no plan-specific pay	

Our Setting: Covered California ACA Exchanges



In all cases, households in a given market and year see the same plans.

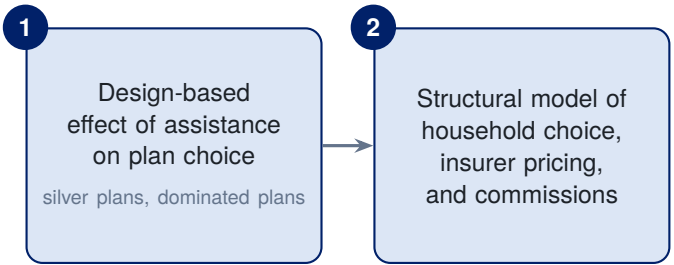
Main elements of the paper

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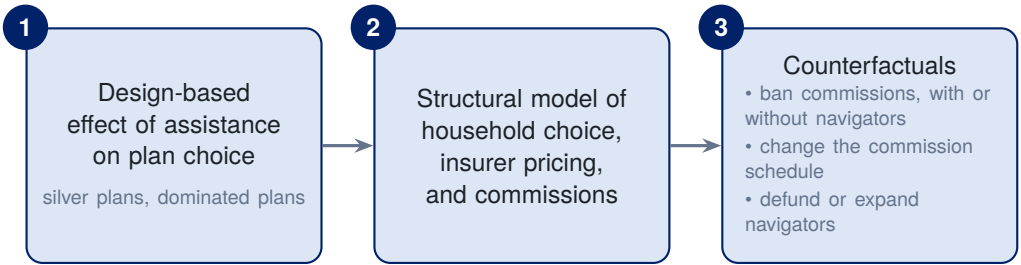
Design-based
effect of assistance
on plan choice

silver plans, dominated plans

Main elements of the paper



Main elements of the paper



Takeaways

1. Restrictions on agent commissions should be evaluated together with what replaces agent assistance. Without a replacement, coverage falls. With navigator assistance in its place, coverage is about unchanged and premiums fall.

2. Changes to how agents are paid have small effects on coverage and household welfare compared with changes in access to assistance.

Contribution

- **Decision assistance and expert advice:** commissioned agents and public navigators compared in the same market
  Schotter (2003) AER; Bettinger et al. (2012) QJE; Kling et al. (2012) QJE; Finkelstein and Notowidigdo (2019) QJE; Bundorf et al. (2024) MS; Aizawa and Kim (2025) AEJ:Pol; Aizawa et al. (WP); Li et al. (WP)

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- **Insurer pricing in the exchanges:** decision assistance and agent commissions added to structural models of exchange pricing
  Starc (2014) RAND; Saltzman (2021) RAND; Saltzman (2025) REStat; Tebaldi (2025) ReStud; Egan et al. (2019) JPE

Setting and Data

Covered California, 2014–2019

- California's ACA marketplace: about one million enrolled households a year, 10–12 insurers, 19 rating regions
- Four metal tiers with standardized cost sharing
- Premiums vary only by age and region, with subsidies up to 400% of FPL (set by the second-cheapest silver plan)
- Cost-sharing subsidies for households below 250% of FPL (available only for silver plans)
- Average net premium of around $130 per member per month

premiums and subsidies

Types of Assistance

Agents and brokers

- Licensed and certified by the exchange
- Paid by insurers, as a flat amount per member or a share of premium, about $13.17 per member per month on average

Types of Assistance

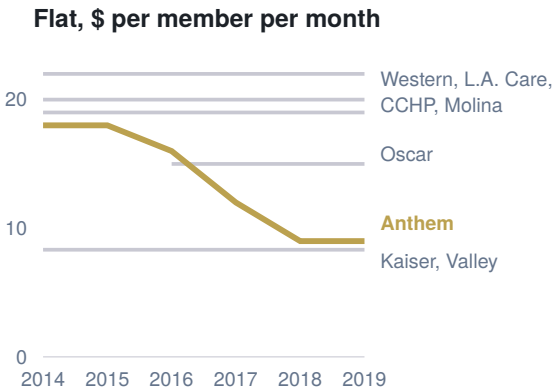
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Navigators

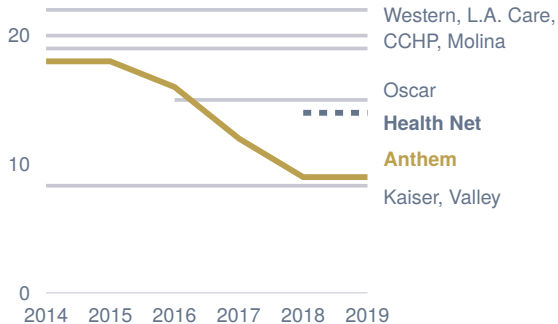
- Certified enrollment counselors and service-center staff
- Funded by the exchange and paid regardless of plan choices

Variation in Commissions

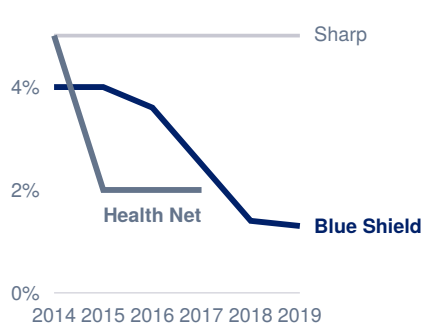


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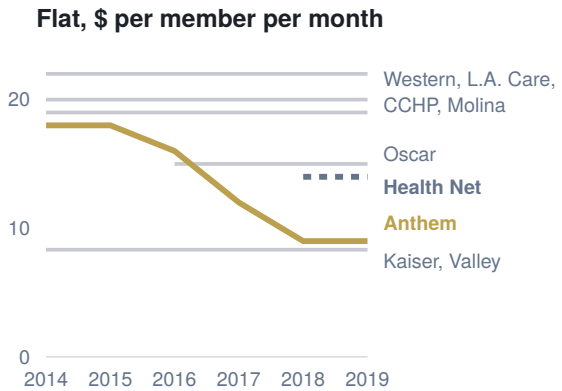
Flat, $ per member per month

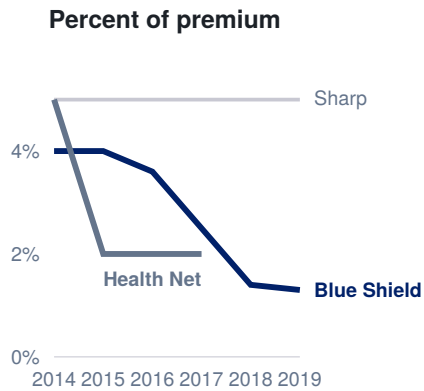


Percent of premium



Variation in Commissions





Only Anthem, Blue Shield, and Health Net change commissions over time.

Measuring the uninsured

- We observe a household's exchange enrollment in every year it is enrolled



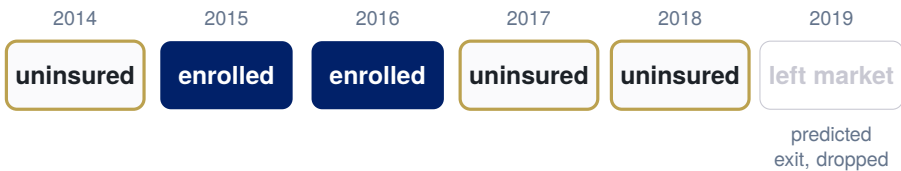
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- Assumed uninsured otherwise, provided it is still eligible for the individual market (following Saltzman 2025, *REStat*)
- Eligibility is estimated based on data from the SIPP (employer coverage, moving, Medicare at 65, Medicaid) using age, income, gender, race, and household size



Estimation sample

Observed: 5,396,359 enrolled household-years

- Plan, ages, income, region, and assistance channel
- Plan menus, premiums, and commission schedules
- Rate filings and MLR filings

With constructed uninsured years:
7,580,377 household-years

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Enrolled household-years

	Assisted	Unassisted
Household-years	3.48M	1.92M
Income (% of FPL)	222	238
Age 18–34	26%	37%
Age 55–64	33%	24%
Dominated choice	2.2%	3.9%

Does Assistance Improve Choices?

Reduced-form design

1. Propensity of assistance from household characteristics

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2. Choice model fit on unassisted households, reweighted by $\hat{p}/(1-\hat{p})$

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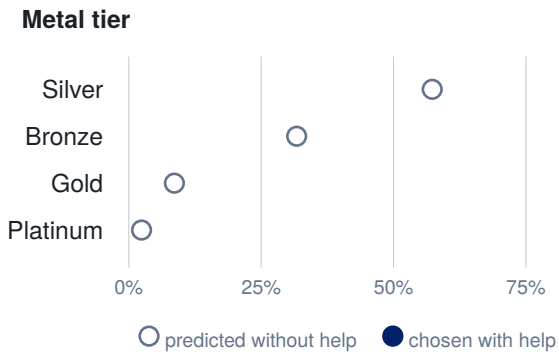
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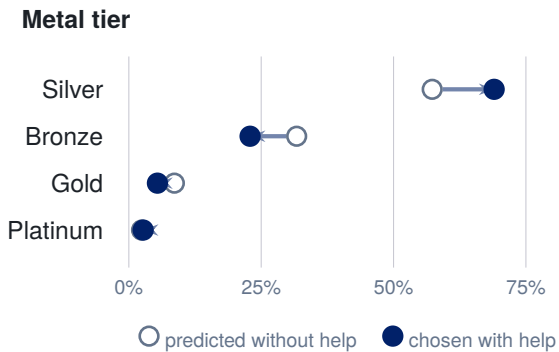
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Outcomes are metal tier, insurer, and dominated-plan choice, with standard errors bootstrapped within region-year cells.

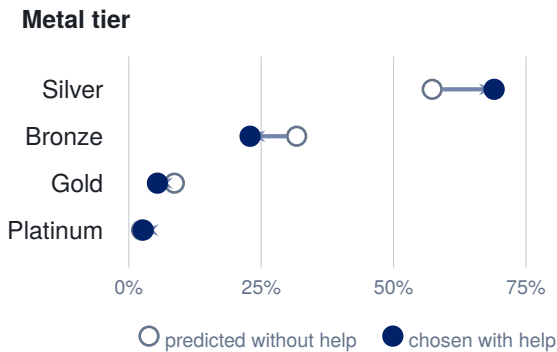
Effects of assistance on plan choice

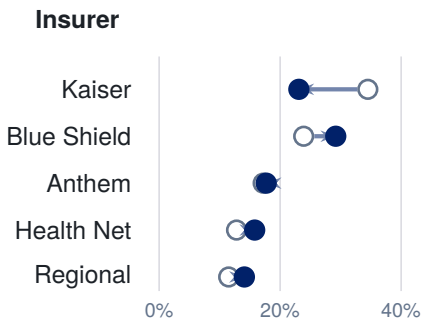


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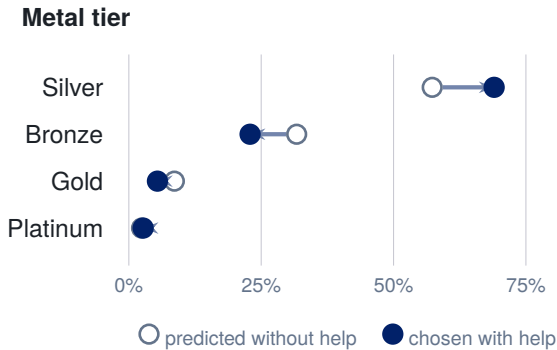


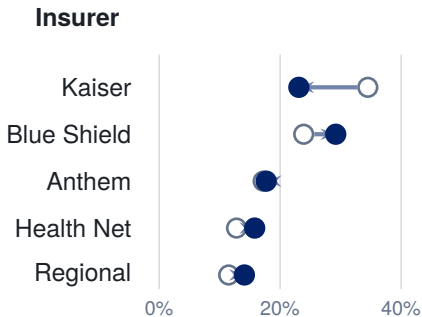
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Effects of assistance on plan choice





Kaiser pays the lowest commission of the large insurers, a flat $6.25 per member per month.

Dominated plans

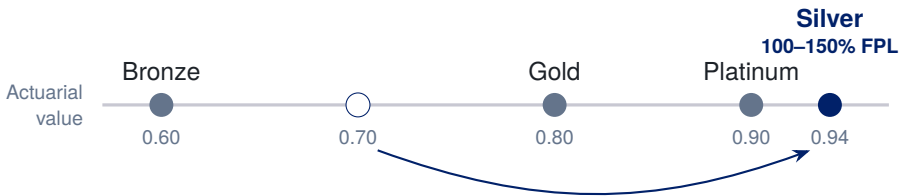


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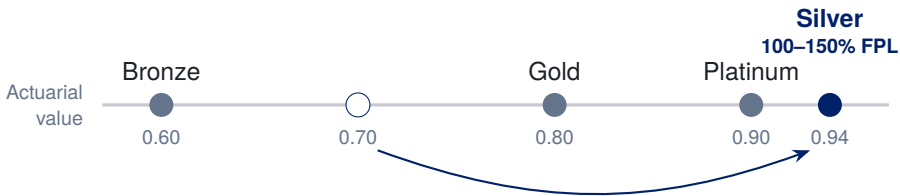
Silver-tier plans are both more generous and cheaper.

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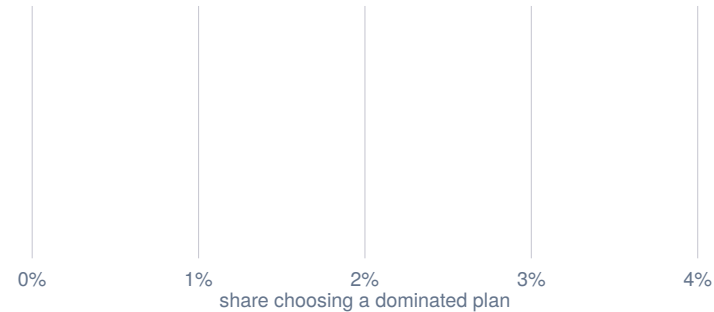
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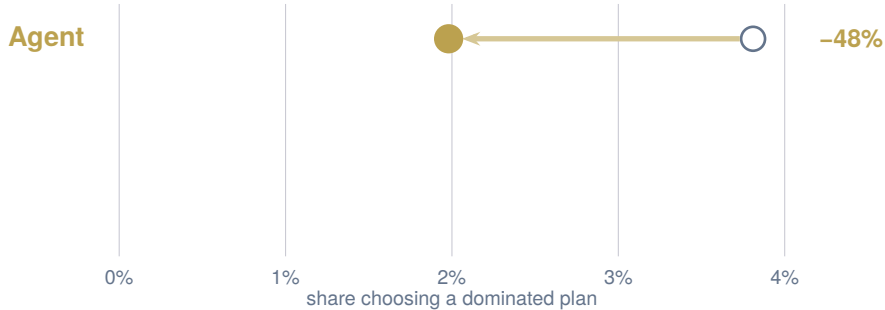
Silver-tier plans are both more generous and cheaper.

Gold and platinum plans are strictly dominated by silver.

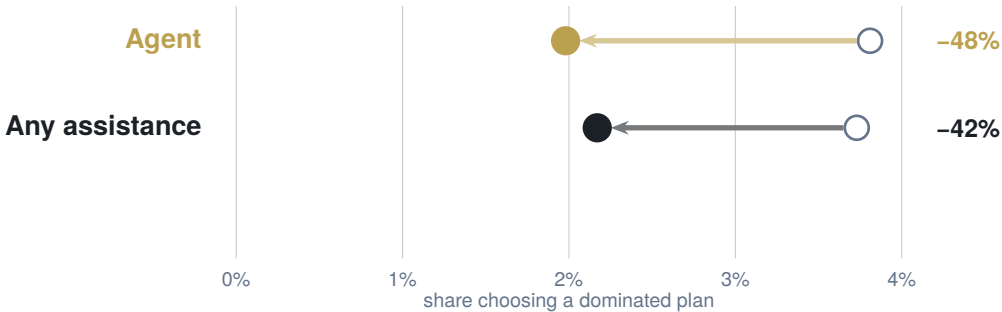
Effects of assistance on dominated choices



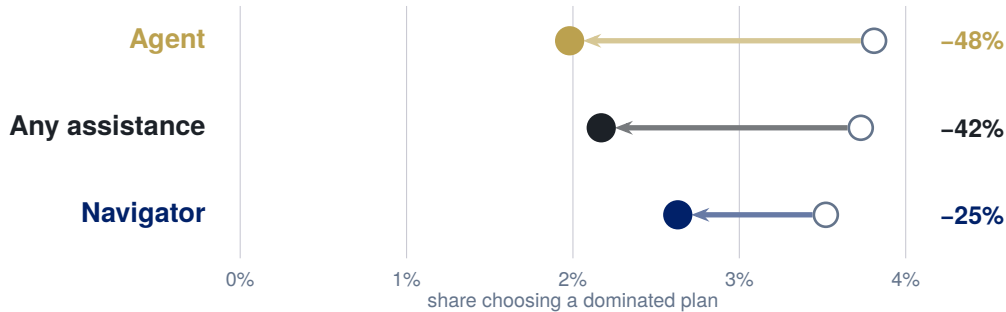
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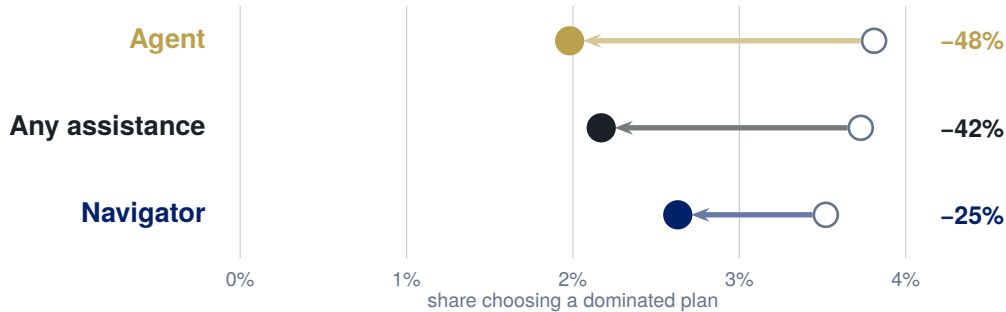
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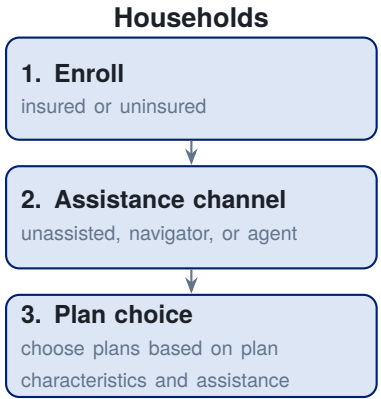
Effects of assistance on dominated choices



Slightly larger effects among new enrollees (2.3pp)

A Model of Choice, Pricing, and Commissions

Basic Setup


Households
1. Enroll
insured or uninsured
2. Assistance channel
unassisted, navigator, or agent
3. Plan choice
choose plans based on plan characteristics and assistance

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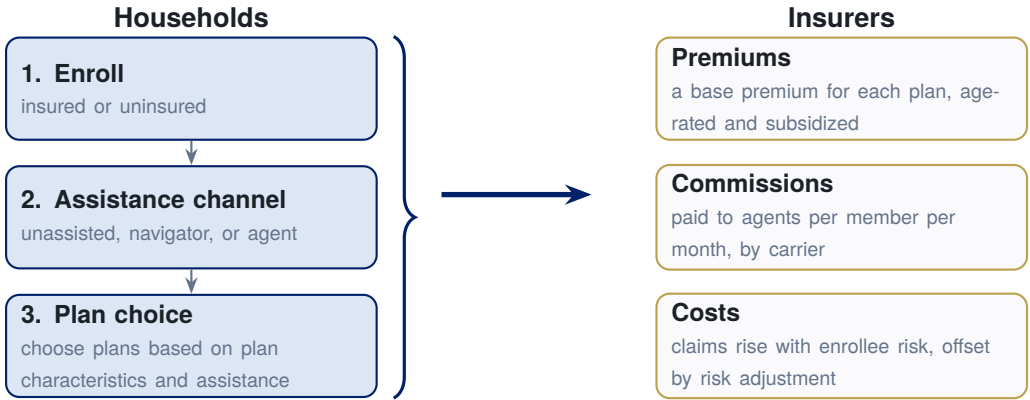
Insurers

Premiums
a base premium for each plan, age-rated and subsidized

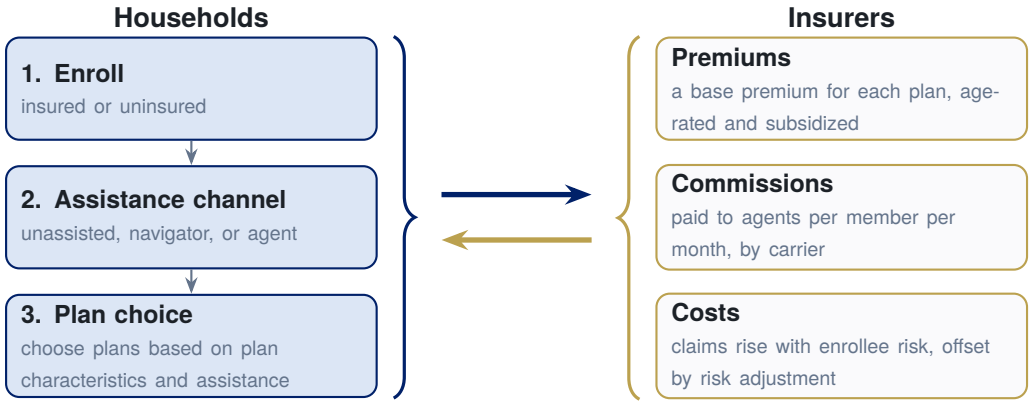
Commissions
paid to agents per member per month, by carrier

Costs
claims rise with enrollee risk, offset by risk adjustment

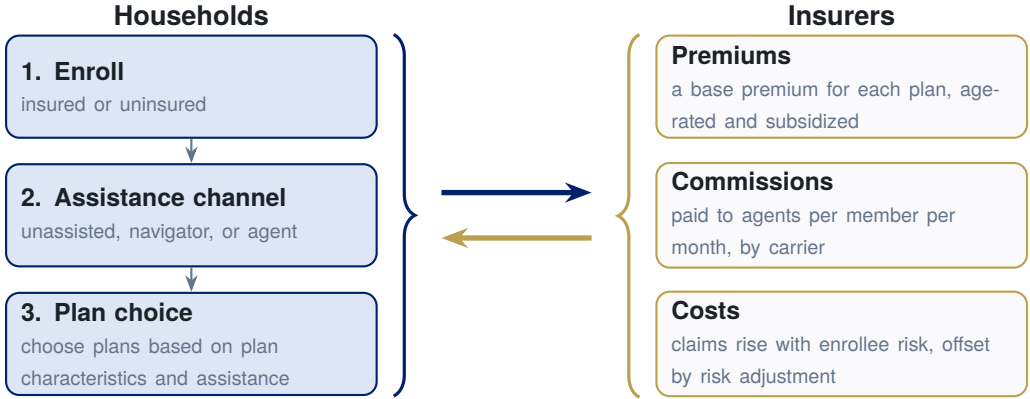
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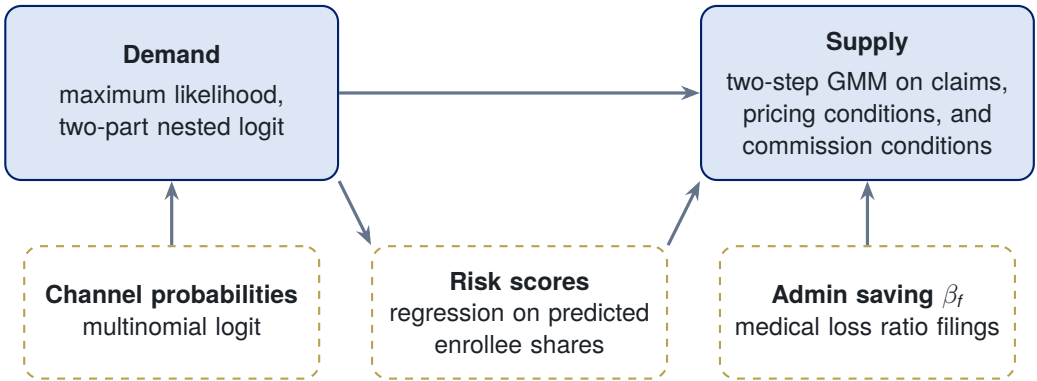


Insurers set premiums and commissions in a Bertrand-Nash equilibrium.

Estimation sequence



Estimation sequence



Household Choice

Stage 3 **Plan choice given the assistance channel**

$$U^A_{ijt} = \overbrace{\delta_0 + \delta' w_{it}}^{\text{coverage}} + \overbrace{\alpha_{it} p_{ijt}(\mathbf{p})}^{\text{net premium}} + \overbrace{\beta^{AV}_{it} AV_{ij}}^{\text{generosity}} + \overbrace{x'_j \beta^x}^{\text{insurer}}$$
$$+ \overbrace{\beta^A AV_{ij} + \phi^A p_{ijt}}^{\text{channel}} + \overbrace{\beta^y_1 y_{jt} + \beta^y_2 y^2_{jt}}^{\text{commission}} + \eta_{it} + \varepsilon_{ijt}$$

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Insured vs uninsured

IN THE MODEL

Household-specific intercept for every exchange plan

SOURCE

Household size, ages, gender, income, and race and ethnicity from the enrollment records (imputed for uninsured years)

Household Choice

Stage 3 **Plan choice given the assistance channel**

$$U_{ijt}^{A} = \overbrace{\delta_0 + \delta' w_{it}}^{\text{coverage}} + \overbrace{\alpha_{it}\, p_{ijt}(\mathbf{p})}^{\text{net premium}} + \overbrace{\beta_{it}^{AV} AV_{ij}}^{\text{generosity}} + \overbrace{x_j' \beta^{x}}^{\text{insurer}} + \overbrace{\beta^{A} AV_{ij} + \phi^{A} p_{ijt}}^{\text{channel}} + \overbrace{\beta_1^{y} y_{jt} + \beta_2^{y} y_{jt}^2}^{\text{commission}} + \eta_{it} + \varepsilon_{ijt}$$

Premium paid

IN THE MODEL

Age-rated premium less the subsidy (based on income and benchmark silver plan)

SOURCE

Base premiums from Covered California's plan data, the ACA age rating curve, IRS contribution percentages, and HHS poverty guidelines

Household Choice

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Plan generosity

IN THE MODEL

Actuarial value, adjusted for the household's cost-sharing reduction

SOURCE

Metal tier from the plan data, and household income from the enrollment records (forms cost-sharing subsidy for silver plans)

Household Choice

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Insurer and network

IN THE MODEL

HMO indicator and brand indicators for the four largest insurers

SOURCE

Insurer and network type from Covered California's plan data

Household Choice

Stage 3 **Plan choice given the assistance channel**

$$U_{ijt}^{A} = \overbrace{\delta_0 + \delta' w_{it}}^{\text{coverage}} + \overbrace{\alpha_{it} p_{ijt}(\mathbf{p})}^{\text{net premium}} + \overbrace{\beta_{it}^{AV} AV_{ij}}^{\text{generosity}} + \overbrace{x_j' \beta^{x}}^{\text{insurer}}$$
$$+ \overbrace{\beta^{A} AV_{ij} + \phi^{A} p_{ijt}}^{\text{channel}} + \overbrace{\beta_1^{y} y_{jt} + \beta_2^{y} y_{jt}^2}^{\text{commission}} + \eta_{it} + \varepsilon_{ijt}$$

Assistance

IN THE MODEL

Assistance changes household responsivness to generosity and premiums

SOURCE

Assistance channel recorded for each enrollment record in the Covered California data

Household Choice

Stage 3 **Plan choice given the assistance channel**

$$U^A_{ijt} = \overbrace{\delta_0 + \delta' w_{it}}^{\text{coverage}} + \overbrace{\alpha_{it}\, p_{ijt}(\mathbf{p})}^{\text{net premium}} + \overbrace{\beta^{AV}_{it} AV_{ij}}^{\text{generosity}} + \overbrace{x_j' \beta^x}^{\text{insurer}}$$
$$+ \overbrace{\beta^A AV_{ij} + \phi^A p_{ijt}}^{\text{channel}} + \overbrace{\beta^y_1 y_{jt} + \beta^y_2 y^2_{jt}}^{\text{commission}} + \eta_{it} + \varepsilon_{ijt}$$

Commission steering

IN THE MODEL

Agents influence choice directly via commissions, with diminishing returns

SOURCE

Commission schedules published by Covered California for each insurer and year, converted to dollars per member per month

Household Choice

Stage 3 **Plan choice given the assistance channel**

$$U^A_{ijt} = \overbrace{\delta_0 + \delta' w_{it}}^{\text{coverage}} + \overbrace{\alpha_{it}\, p_{ijt}(\mathbf{p})}^{\text{net premium}} + \overbrace{\beta^{AV}_{it} AV_{ij}}^{\text{generosity}} + \overbrace{x_j' \beta^x}^{\text{insurer}}$$
$$+ \overbrace{\beta^A AV_{ij} + \phi^A p_{ijt}}^{\text{channel}} + \overbrace{\beta^y_1 y_{jt} + \beta^y_2 y^2_{jt}}^{\text{commission}} + \eta_{it} + \varepsilon_{ijt}$$

The other two channels have the following utility.

- Unassisted (U^0_{ijt}) utility includes the coverage, premium, generosity, and insurer terms only
- Navigator (U^N_{ijt}) utility adds its own channel terms, $\beta^N AV_{ij} + \phi^N p_{ijt}$, and no commission terms

Household Choice

Stage 3 **Plan choice given the assistance channel**

$$U^c_{ijt} = \delta_0 + \delta' w_{it} + \overbrace{V^c_{ijt}}^{\text{varies by plan}} + \overbrace{\eta_{it} + \varepsilon_{ijt}}^{\text{shocks}}$$

- η_{it} is unobserved (to econometrician) taste for insurance coverage
- ε_{ijt} is a plan shock, type I extreme value with scale λ, drawn when the household compares plans

Household Choice

Stage 3 **Plan choice given the assistance channel**

$$U^c_{ijt} = \delta_0 + \delta' w_{it} + \overbrace{V^c_{ijt}}^{\text{varies by plan}} + \overbrace{\eta_{it} + \varepsilon_{ijt}}^{\text{shocks}}$$

$$\Pr(j \mid c) = \frac{e^{V^c_{ijt}/\lambda}}{\sum_k e^{V^c_{ikt}/\lambda}}$$

- η_{it} and the coverage term $\delta_0 + \delta' w_{it}$ cancel since they are the same for every plan

Household Choice

Stage 3 Plan choice given the channel $e^{V^c_{ijt}/\lambda} \Big/ \sum_k e^{V^c_{ikt}/\lambda}$

Stage 2 **Average expected value of plans across assistance channels**

$$\mathrm{E}\left[\max_k U^c_{ikt} \mid c, \eta_{it}\right] = \delta_0 + \delta' w_{it} + \eta_{it} + \lambda I^c_{it}, \qquad I^c_{it} = \ln \sum_k e^{V^c_{ikt}/\lambda}$$

$$\bar{I}_{it} = \sum_c \overbrace{\pi_{ict}}^{\text{channel probability}} I^c_{it}$$

- The household can anticipate the expected value of plans for each assistance channel, but not yet how or if it will receive assistance

Household Choice

Stage 3 Plan choice given the channel $e^{V^c_{ijt}/\lambda} \Big/ \sum_k e^{V^c_{ikt}/\lambda}$

Stage 2 **Average expected value of plans across assistance channels**

$$\mathrm{E}\left[\max_k U^c_{ikt} \mid c, \eta_{it}\right] = \delta_0 + \delta' w_{it} + \eta_{it} + \lambda I^c_{it}, \qquad I^c_{it} = \ln \sum_k e^{V^c_{ikt}/\lambda}$$

$$\bar{I}_{it} = \sum_c \overbrace{\pi_{ict}}^{\text{channel probability}} I^c_{it}$$

Assistance channel probabilities

IN THE MODEL

Multinomial logit on demographics, agent density, and year effects

SOURCE

Enrollment records, Covered California agent counts by region, and Census population

Household Choice

Stage 3 Plan choice given the channel $e^{V^c_{ijt}/\lambda} \Big/ \sum_k e^{V^c_{ikt}/\lambda}$

Stage 2 Expected value across channels $\bar{I}_{it} = \sum_c \pi_{ict} I^c_{it}$

Stage 1 **Insured or uninsured**

$$\text{enroll if } \overbrace{\delta_0 + \delta' w_{it} + \lambda \bar{I}_{it}}^{\text{value of enrolling } V^E_{it}} + \eta_{it} \geq \overbrace{\alpha_{it}\, \rho_{it}}^{\text{outside option } V_{i0t}} + \varepsilon_{i0t}$$

- The household knows η_{it} but not yet its channel or its plan shocks

Household Choice

Stage 3 Plan choice given the channel $e^{V^c_{ijt}/\lambda} \Big/ \sum_k e^{V^c_{ikt}/\lambda}$

Stage 2 Expected value across channels $\bar{I}_{it} = \sum_c \pi_{ict} I^c_{it}$

Stage 1 **Insured or uninsured**

$$\text{enroll if } \overbrace{\delta_0 + \delta' \boldsymbol{w}_{it} + \lambda \bar{I}_{it}}^{\text{value of enrolling } V^E_{it}} + \eta_{it} \geq \overbrace{\alpha_{it}\, \rho_{it}}^{\text{outside option } V_{i0t}} + \varepsilon_{i0t}$$

- The expected maximum value of plan options, averaged over channels, plus the coverage term

Household Choice

Stage 3 Plan choice given the channel $e^{V^c_{ijt}/\lambda} \Big/ \sum_k e^{V^c_{ikt}/\lambda}$

Stage 2 Expected value across channels $\bar{I}_{it} = \sum_c \pi_{ict} I^c_{it}$

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Mandate penalty

IN THE MODEL

Utility of going uninsured, the premium coefficient times the penalty ρ_{it}

SOURCE

IRS penalty schedule, zero in 2019, applied to income and household size from the enrollment records

Household Choice

Stage 3 Plan choice given the channel $e^{V^c_{ijt}/\lambda} \Big/ \sum_k e^{V^c_{ikt}/\lambda}$

Stage 2 Expected value across channels $\bar{I}_{it} = \sum_c \pi_{ict} I^c_{it}$

Stage 1 **Insured or uninsured**

$$\text{enroll if } \overbrace{\delta_0 + \delta' \boldsymbol{w}_{it} + \lambda \bar{I}_{it}}^{\text{value of enrolling } V^E_{it}} + \eta_{it} \geq \overbrace{\alpha_{it}\, \rho_{it}}^{\text{outside option } V_{i0t}} + \varepsilon_{i0t}$$

$$\Pr(\text{enroll}) = \frac{e^{V^E_{it}}}{e^{V^E_{it}} + e^{V_{i0t}}}$$

- η_{it} and ε_{i0t} are independent type I extreme value, so the enrollment decision is a binary logit

Household Choice

Stage 3	Plan choice given the channel	$e^{V^c_{ijt}/\lambda} \big/ \sum_k e^{V^c_{ikt}/\lambda}$
Stage 2	Expected value across channels	$\bar{I}_{it} = \sum_c \pi_{ict} I^c_{it}$
Stage 1	Insured or uninsured	$e^{V^E_{it}} \big/ \left(e^{V^E_{it}} + e^{V_{i0t}}\right)$

$$q_{ijt} = \underbrace{\frac{e^{V_{ijt}/\lambda}}{\sum_k e^{V_{ikt}/\lambda}}}_{\text{plan, at the realized channel}} \times \underbrace{\frac{e^{V^E_{it}}}{e^{V^E_{it}} + e^{V_{i0t}}}}_{\text{enrollment}}$$

- With a single channel, this is the standard nested logit with nesting parameter λ

Estimating demand

1. Channel probabilities π_{ict}: multinomial logit on enrolled households, with demographics, licensed agents per 10,000 residents, and year effects

identification · three-level nest · plan-based enrollers · gatekeeper simulation

Estimating demand

1. Channel probabilities π_{ict}: multinomial logit on enrolled households, with demographics, licensed agents per 10,000 residents, and year effects
2. Maximum likelihood on q_{ijt}, taking π_{ict} as given, with plan choices from enrolled household-years and the enrollment decision from enrolled and constructed uninsured household-years

identification | three-level nest | plan-based enrollers | gatekeeper simulation

Estimating demand

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3. A 20% random sample of households within each rating region and year (1,492,674 household-years), weighted by household size

identification | three-level nest | plan-based enrollers | gatekeeper simulation

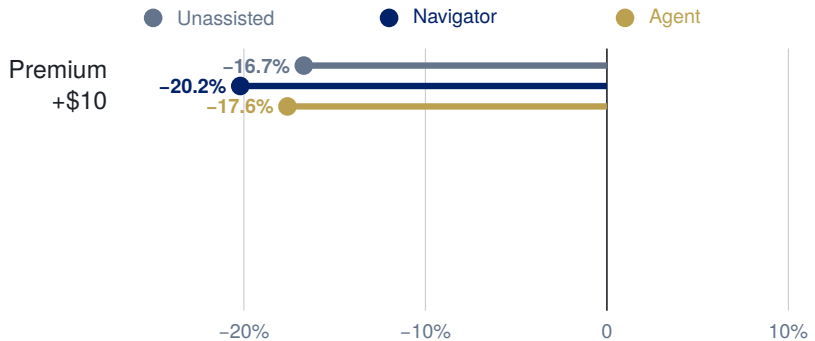
Estimating demand

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2. Maximum likelihood on q_{ijt}, taking π_{ict} as given, with plan choices from enrolled household-years and the enrollment decision from enrolled and constructed uninsured household-years
3. A 20% random sample of households within each rating region and year (1,492,674 household-years), weighted by household size

48 parameters, including λ, with robust standard errors

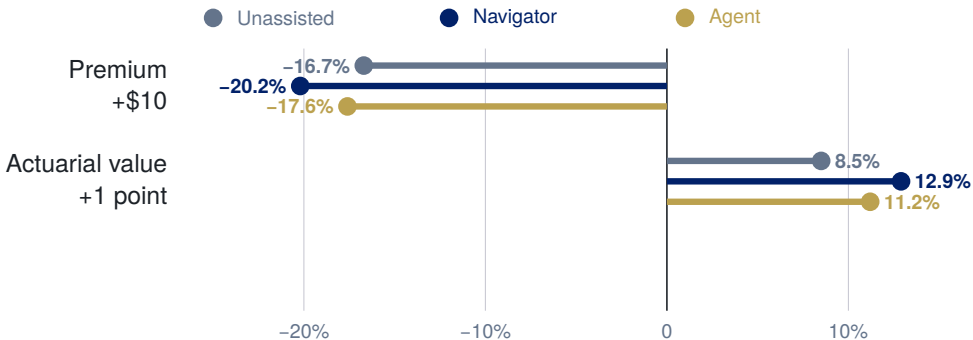
identification | three-level nest | plan-based enrollers | gatekeeper simulation

Results for demand estimation



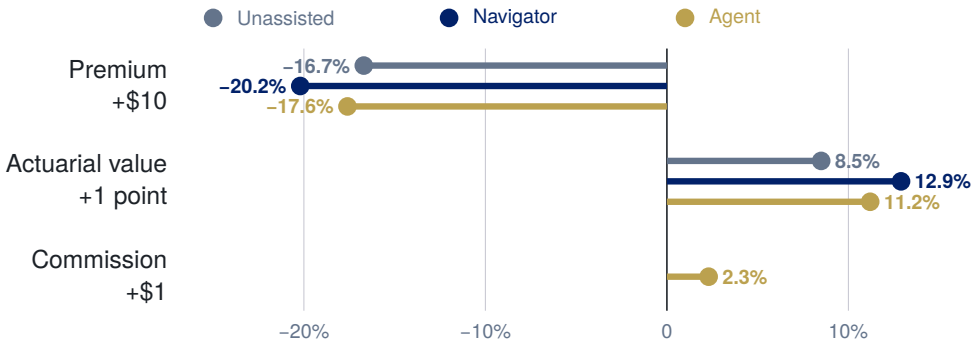
Each bar is the percent change in a plan's choice probability when that plan's premium, actuarial value, or commission changes by the stated amount. Premiums and commissions are per member per month.

Results for demand estimation



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Results for demand estimation



Each bar is the percent change in a plan's choice probability when that plan's premium, actuarial value, or commission changes by the stated amount. Premiums and commissions are per member per month.

Insurer profits

$$\pi_f = \overbrace{R_f}^{\text{premiums}} - \overbrace{C_f}^{\text{claims}} + \overbrace{RA_f}^{\text{risk adjustment}} + \overbrace{RI_f}^{\text{reinsurance}} - \overbrace{Y_f}^{\text{commissions}} - \overbrace{VC_f}^{\text{admin costs}}$$

Insurer profits

$$\pi_f = \overbrace{R_f}^{\text{premiums}} - \overbrace{C_f}^{\text{claims}} + \overbrace{RA_f}^{\text{risk adjustment}} + \overbrace{RI_f}^{\text{reinsurance}} - \overbrace{Y_f}^{\text{commissions}} - \overbrace{VC_f}^{\text{admin costs}}$$

Premium revenue

IN THE MODEL

Base premium times age rating factor, summed over predicted enrollment

SOURCE

Base premiums from Covered California's plan data and the ACA age rating curve

Insurer profits

$$\pi_f = \overbrace{R_f}^{\text{premiums}} - \overbrace{C_f}^{\text{claims}} + \overbrace{RA_f}^{\text{risk adjustment}} + \overbrace{RI_f}^{\text{reinsurance}} - \overbrace{Y_f}^{\text{commissions}} - \overbrace{VC_f}^{\text{admin costs}}$$

Claims

IN THE MODEL

Average claims rise with predicted enrollee risk, so a lower premium that draws in sicker members raises cost

SOURCE

Plan-year claims from the CMS rate filings (URRT), 2014–2018

Insurer profits

$$\pi_f = \overbrace{R_f}^{\text{premiums}} - \overbrace{C_f}^{\text{claims}} + \overbrace{RA_f}^{\text{risk adjustment}} + \overbrace{RI_f}^{\text{reinsurance}} - \overbrace{Y_f}^{\text{commissions}} - \overbrace{VC_f}^{\text{admin costs}}$$

Risk adjustment

IN THE MODEL

Redistributes revenue from plans with healthier enrollees to those with sicker enrollees (”sicker” or ”healthier” relative to age of enrollees)

SOURCE

Plan-year risk scores and transfers from the rate filings (SRRT), and the CMS transfer formula

Insurer profits

$$\pi_f = \overbrace{R_f}^{\text{premiums}} - \overbrace{C_f}^{\text{claims}} + \overbrace{RA_f}^{\text{risk adjustment}} + \overbrace{RI_f}^{\text{reinsurance}} - \overbrace{Y_f}^{\text{commissions}} - \overbrace{VC_f}^{\text{admin costs}}$$

Reinsurance

IN THE MODEL

Federal payments covering part of the claims of very high-cost enrollees

SOURCE

Reinsurance recoveries reported in the rate filings, 2014–2016

Insurer profits

$$\pi_f = \overbrace{R_f}^{\text{premiums}} - \overbrace{C_f}^{\text{claims}} + \overbrace{RA_f}^{\text{risk adjustment}} + \overbrace{RI_f}^{\text{reinsurance}} - \overbrace{Y_f}^{\text{commissions}} - \overbrace{VC_f}^{\text{admin costs}}$$

Commissions

IN THE MODEL

Paying an agent replaces in-house enrollment and servicing, so a commission dollar costs $1 - \beta_f$

SOURCE

Commission schedules from Covered California, and commissions and other administrative expense from the medical loss ratio filings

Costs of Enrollees

$$\ln r_{jt} = \overbrace{\gamma^{f\times m}}^{\text{insurer} \times \text{metal}} + \sum_d \gamma^d \overbrace{s_{djt}(\mathbf{p}, \mathbf{y})}^{\text{enrollee mix from demand}}$$

$$\ln c_{jt} = \overbrace{\mu^r \ln \hat{r}_{jt}}^{\text{risk pass-through}} + x_j' \mu^x + \mu^l l_t$$

We observe risk scores r (by insurer, metal, region, and year) and average claims c (by plan-year, 2014–2018) in the rate filings. Claims depend on the fitted risk score $\hat{r}$, which depends on enrollments.

Costs of Enrollees

$$\ln r_{jt} = \overbrace{\gamma^{f \times m}}^{\text{insurer} \times \text{metal}} + \sum_d \gamma^d \overbrace{s_{djt}(\mathbf{p}, \mathbf{y})}^{\text{enrollee mix from demand}}$$

$$\ln c_{jt} = \overbrace{\mu^r \ln \hat{r}_{jt}}^{\text{risk pass-through}} + x_j' \mu^x + \mu^l l_t$$

Risk level by insurer and metal

IN THE MODEL

A baseline risk level for each insurer's plans in each metal tier

SOURCE

Risk scores from the rate filings (SRRT), by insurer, metal, region, and year

Costs of Enrollees

$$\ln r_{jt} = \overbrace{\gamma^{f\times m}}^{\text{insurer} \times \text{metal}} + \sum_d \gamma^d \overbrace{s_{djt}(\mathbf{p}, \mathbf{y})}^{\text{enrollee mix from demand}}$$

$$\ln c_{jt} = \overbrace{\mu^r \ln \hat{r}_{jt}}^{\text{risk pass-through}} + x_j'\mu^x + \mu^l l_t$$

Enrollee mix

IN THE MODEL

Shares of members aged 0–34, male, in family households, and non-white, from the enrollment the demand model predicts

SOURCE

Household demographics from the enrollment records

Costs of Enrollees

$$\ln r_{jt} = \overbrace{\gamma^{f\times m}}^{\text{insurer} \times \text{metal}} + \sum_d \gamma^d \overbrace{s_{djt}(\mathbf{p}, \mathbf{y})}^{\text{enrollee mix from demand}}$$

$$\ln c_{jt} = \overbrace{\mu^r \ln \hat{r}_{jt}}^{\text{risk pass-through}} + x_j'\mu^x + \mu^l l_t$$

Claims pass-through: 1.31

IN THE MODEL

Claims rise about 13% when the fitted risk score rises 10%

SOURCE

Plan-year claims from the rate filings (URRT), 2014–2018, estimated jointly with the pricing and commission conditions

Pricing FOC

$$\underbrace{\frac{\partial R_f}{\partial p_j} + \frac{\partial RA_f}{\partial p_j} + \frac{\partial RI_f}{\partial p_j}}_{\text{marginal revenue, with transfers}} = \underbrace{\frac{\partial C_f}{\partial p_j}}_{\text{selection}} + \underbrace{\frac{\partial Y_f}{\partial p_j} + \frac{\partial VC_f}{\partial p_j}}_{\text{commissions and admin costs}}$$

model fit

Pricing FOC

$$\underbrace{\frac{\partial R_f}{\partial p_j} + \frac{\partial RA_f}{\partial p_j} + \frac{\partial RI_f}{\partial p_j}}_{\text{marginal revenue, with transfers}} = \underbrace{\frac{\partial C_f}{\partial p_j}}_{\text{selection}} + \underbrace{\frac{\partial Y_f}{\partial p_j} + \frac{\partial VC_f}{\partial p_j}}_{\text{commissions and admin costs}}$$

Revenue from charging current enrollees more, less the revenue lost from enrollees who leave, plus the change in risk adjustment and reinsurance

model fit

Pricing FOC

$$\underbrace{\frac{\partial R_f}{\partial p_j} + \frac{\partial RA_f}{\partial p_j} + \frac{\partial RI_f}{\partial p_j}}_{\text{marginal revenue, with transfers}} = \underbrace{\frac{\partial C_f}{\partial p_j}}_{\text{selection}} + \underbrace{\frac{\partial Y_f}{\partial p_j} + \frac{\partial VC_f}{\partial p_j}}_{\text{commissions and admin costs}}$$

Healthier enrollees are the first to leave when a premium rises, which raises average claims among those who stay.

model fit

Pricing FOC

$$\underbrace{\frac{\partial R_f}{\partial p_j} + \frac{\partial RA_f}{\partial p_j} + \frac{\partial RI_f}{\partial p_j}}_{\text{marginal revenue, with transfers}} = \underbrace{\frac{\partial C_f}{\partial p_j}}_{\text{selection}} + \underbrace{\frac{\partial Y_f}{\partial p_j} + \frac{\partial VC_f}{\partial p_j}}_{\text{commissions and admin costs}}$$

A higher premium loses some enrollees, which lowers the commissions and administrative costs paid by the insurer.

model fit

Commission FOC

$$\underbrace{\sum_k (p_k - \overbrace{mc_k}^{\text{marginal cost}}) \frac{\partial q_k}{\partial y_j}}_{\text{marginal profit from steering}} = \underbrace{\left[(1-\beta_f)+\delta_f\right]}_{\text{net cost per dollar}} \times \frac{\partial Y_f}{\partial y_j}$$

One condition per carrier-year, or carrier-network where HMO and PPO rates differ

model fit

Commission FOC

$$\underbrace{\sum_k (p_k - \overbrace{mc_k}^{\text{marginal cost}}) \frac{\partial q_k}{\partial y_j}}_{\text{marginal profit from steering}} = \underbrace{\left[(1 - \beta_f) + \delta_f\right]}_{\text{net cost per dollar}} \times \frac{\partial Y_f}{\partial y_j}$$

Profit on enrollees gained or lost from a change in commission, with marginal cost net of risk adjustment

model fit

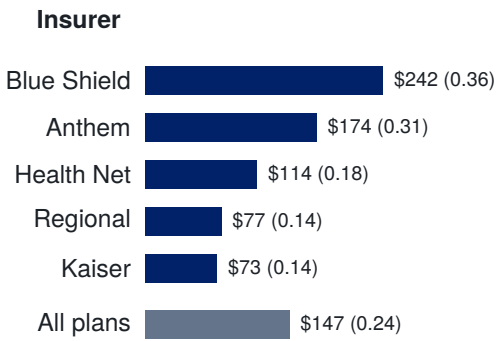
Commission FOC

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- β_f is the in-house enrollment and servicing cost saved per commission dollar, estimated from the medical loss ratio filings
- δ_f is the part of a commission dollar's cost to carrier *f* that steering and administrative savings do not explain, estimated for each carrier
- $\delta_f > 0$ when a carrier pays less than steering alone would justify, largest for Kaiser, which pays the lowest commission

model fit

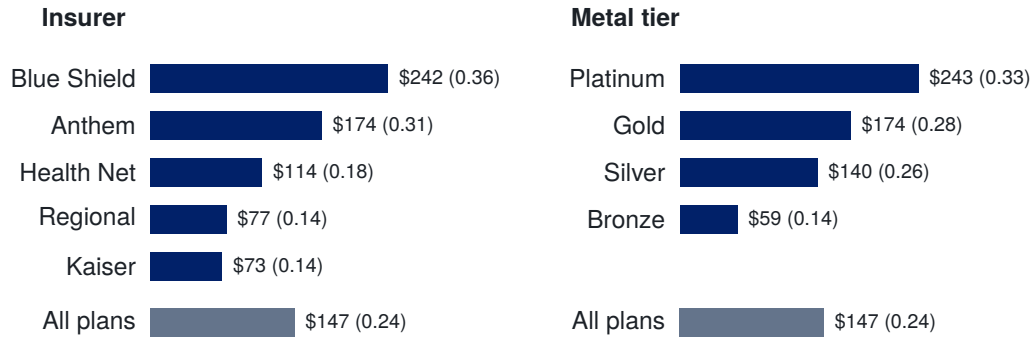
Markups



Markup is the premium less marginal cost (claims net of risk adjustment and reinsurance, plus administrative cost), in dollars per member per month, with the Lerner index in parentheses. Values are means across plans and region-years, 2014–2018.

open questions

Markups



Markup is the premium less marginal cost (claims net of risk adjustment and reinsurance, plus administrative cost), in dollars per member per month, with the Lerner index in parentheses. Values are means across plans and region-years, 2014–2018.

open questions

Welfare Effects of Agents

Welfare Measures

Consumer Surplus

$$CS_i = \frac{1}{\alpha_i}\mathbb{E}\left[\max_j U_{ij}\right] = \frac{1}{\alpha_i}\ln\left(\overbrace{e^{V_{i0}}}^{\text{uninsured}} + \overbrace{e^{V_i^E}}^{\text{enrolling}}\right)$$

- Standard consumer surplus, with commission terms set to zero

Welfare Measures

Consumer Surplus

$$CS_i = \frac{1}{\alpha_i}\mathbb{E}\left[\max_j U_{ij}\right] = \frac{1}{\alpha_i}\ln\left(\overbrace{e^{V_{i0}}}^{\text{uninsured}} + \overbrace{e^{V_i^E}}^{\text{enrolling}}\right)$$

- Standard consumer surplus, with commission terms set to zero
- Problematic with steering, since the agent channel's weights on premiums and generosity count as household preferences

Welfare Measures

Consumer Surplus $CS_i = \frac{1}{\alpha_i} \ln\left(e^{V_{i0}} + e^{V_i^E}\right)$

Consumer Surplus with Navigator Assistance

$$V_i^{nav} = \frac{1}{\alpha_i^{nav}} \sum_j q_{ijt} \, V_{ij}^{nav}$$

- Same choices q_{ijt}, valued with a navigator client's utility
- Systematic part only, with no agent or commission terms

Welfare Measures

Consumer Surplus $CS_i = \frac{1}{\alpha_i} \ln\left(e^{V_{i0}} + e^{V_i^E}\right)$

Consumer Surplus with Navigator Assistance $V_i^{nav} = \frac{1}{\alpha_i^{nav}} \sum_j q_{ijt} V_{ij}^{nav}$

Expected Costs

$$V_i^{obj} = \sum_j q_{ijt} V_{ij}^{obj}, \qquad V_{ij}^{obj} = -\Big(\overbrace{\text{premium}_{ij}}^{\text{premium}} + \overbrace{\mathbb{E}[OOP_{ij}]}^{\text{expected out-of-pocket}} + \overbrace{\tfrac{\rho}{2}\text{Var}[OOP_{ij}]}^{\text{variance charge}}\Big)$$

- Same choices q_{ijt}, valued at expected cost in dollars
- Out-of-pocket spending from each plan's deductible, coinsurance, and out-of-pocket maximum, applied to 2018 MEPS spending by age and income

Welfare Measures

Consumer Surplus $CS_i = \frac{1}{\alpha_i} \ln\left(e^{V_{i0}} + e^{V_i^E}\right)$

Consumer Surplus with Navigator Assistance $V_i^{nav} = \frac{1}{\alpha_i^{nav}} \sum_j q_{ijt} V_{ij}^{nav}$

Expected Costs

$$\text{uninsured:} \quad V_{i0}^{obj} = -\Big(\overbrace{OOP_i^u}^{\text{out-of-pocket}} + \overbrace{RP + \Delta m_i\, VSL + D_i}^{\text{risk protection, mortality, distress}} \Big)$$

- The out-of-pocket amount uninsured households actually pay (2018 MEPS), less than their full medical bill
- Plus three costs of lacking coverage: lost risk protection, a mortality cost (VSL \$13M), and financial distress above 40% of income

Welfare Measures

Consumer Surplus $CS_i = \frac{1}{\alpha_i} \ln \left(e^{V_{i0}} + e^{V_i^E} \right)$

Consumer Surplus with Navigator Assistance $V_i^{nav} = \frac{1}{\alpha_i^{nav}} \sum_j q_{ijt} V_{ij}^{nav}$

Expected Costs $V_i^{obj} = \sum_j q_{ijt} V_{ij}^{obj}$

Producer Surplus

$$PS = \sum_f \left[\sum_{j \in f} (p_j - mc_j)\, q_j - (1 - \beta_f)\, Y_f \right]$$

- Premium revenue less marginal cost (claims net of risk adjustment and reinsurance, plus admin costs)
- Less the net cost of commissions

Welfare Measures

Consumer Surplus $CS_i = \frac{1}{\alpha_i} \ln\left(e^{V_{i0}} + e^{V_i^E}\right)$

Consumer Surplus with Navigator Assistance $V_i^{nav} = \frac{1}{\alpha_i^{nav}} \sum_j q_{ijt} V_{ij}^{nav}$

Expected Costs $V_i^{obj} = \sum_j q_{ijt} V_{ij}^{obj}$

Producer Surplus $PS = \sum_f \left[\sum_{j \in f} (p_j - mc_j)\, q_j - (1 - \beta_f) Y_f \right]$

Government Budget

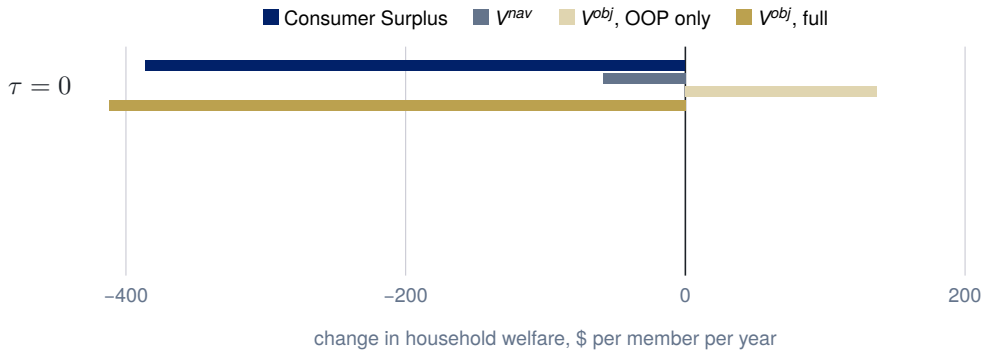
$$G = -\big(\text{tax credits} + \text{CSR payments} + \text{uncompensated care} - \text{penalties}\big)$$

- Tax credits and cost-sharing reduction payments for enrollees
- Uncompensated care: \$2,025 per uninsured person in 2013 (Coughlin et al., 2014), adjusted by year and for the risk of the uninsured pool

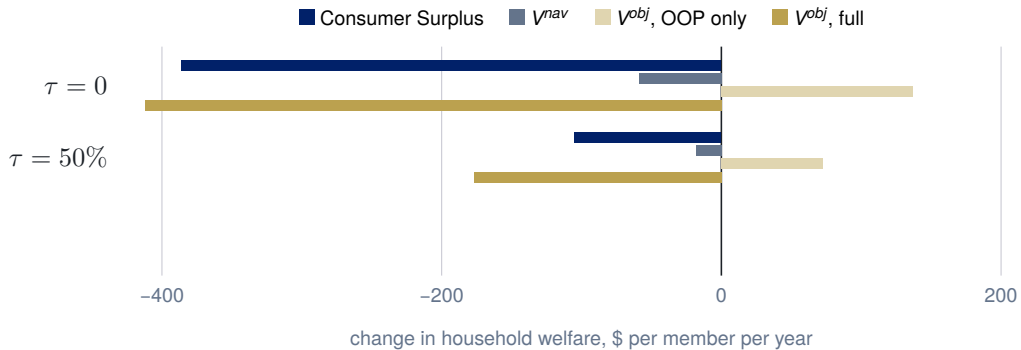
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Consumer Surplus	$CS_i = \frac{1}{\alpha_i} \ln\left(e^{V_{i0}} + e^{V_i^E}\right)$
Consumer Surplus with Navigator Assistance	$V_i^{nav} = \frac{1}{\alpha_i^{nav}} \sum_j q_{ijt} V_{ij}^{nav}$
Expected Costs	$V_i^{obj} = \sum_j q_{ijt} V_{ij}^{obj}$
Producer Surplus	$PS = \sum_f \left[\sum_{j \in f} (p_j - mc_j)\, q_j - (1 - \beta_f) Y_f \right]$
Government Budget	$G = -(\text{tax credits} + \text{CSR} + \text{uncomp. care} - \text{penalties})$

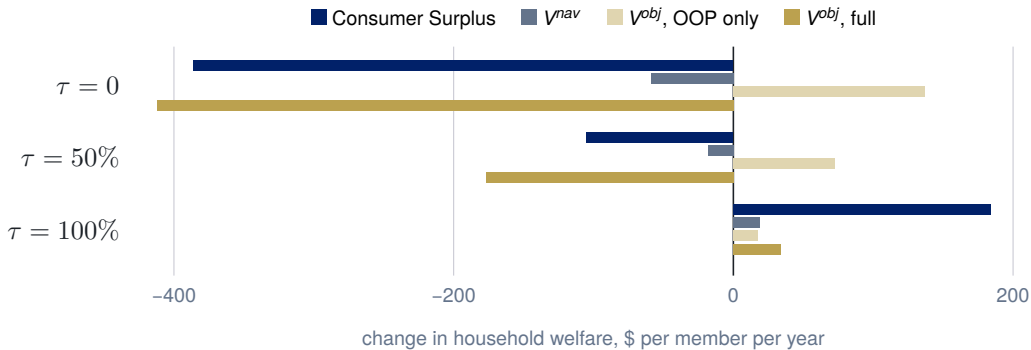
Counterfactual 1: Banning Commissions



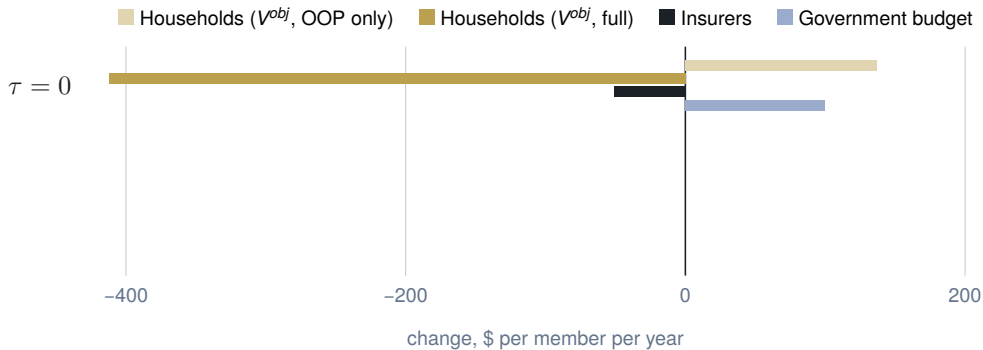
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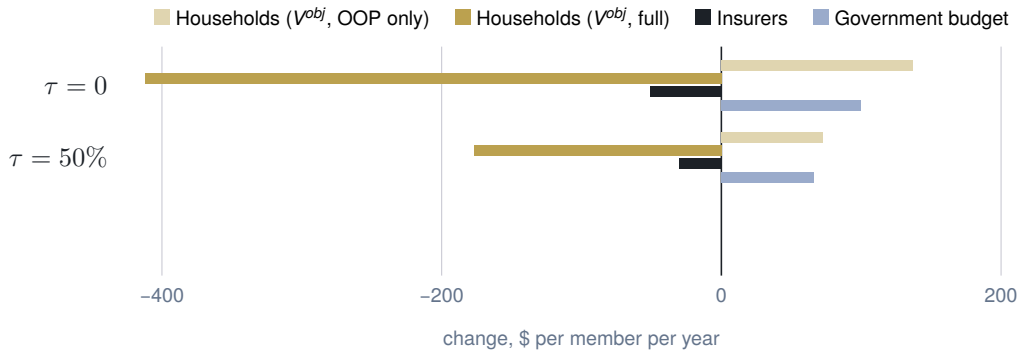
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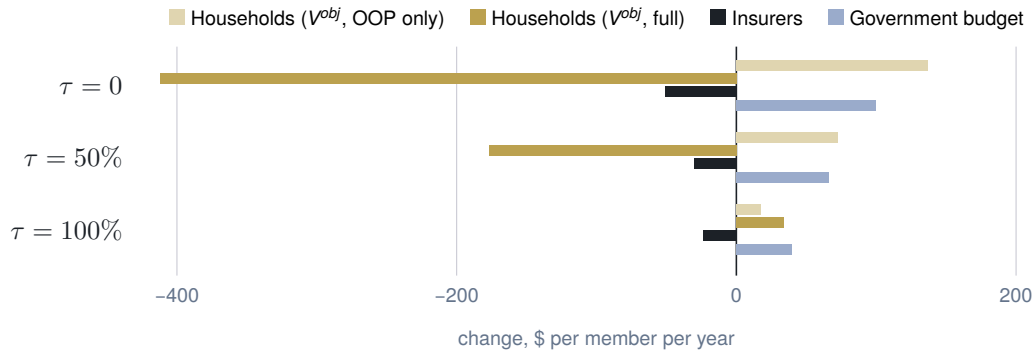
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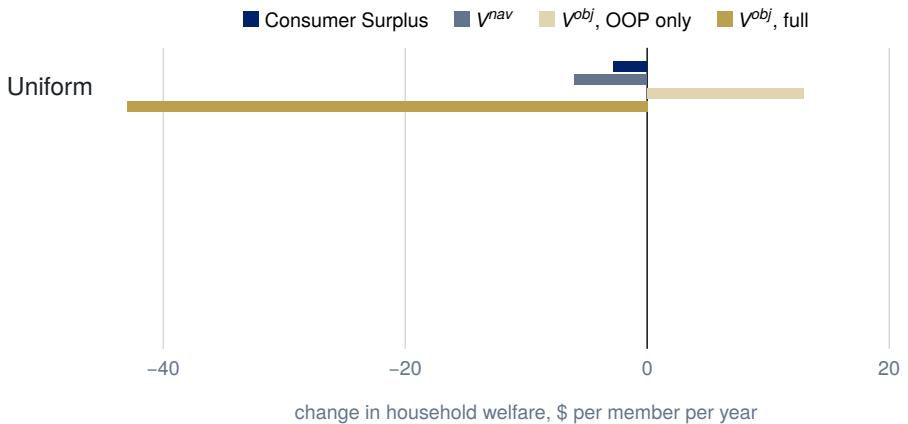
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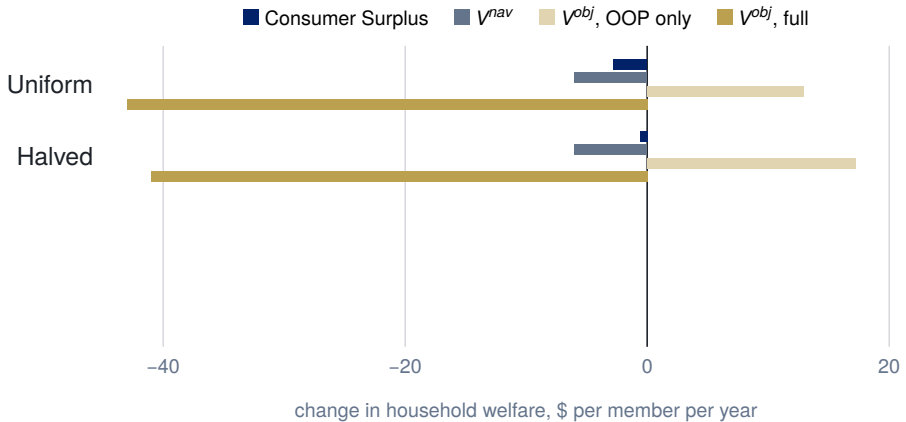
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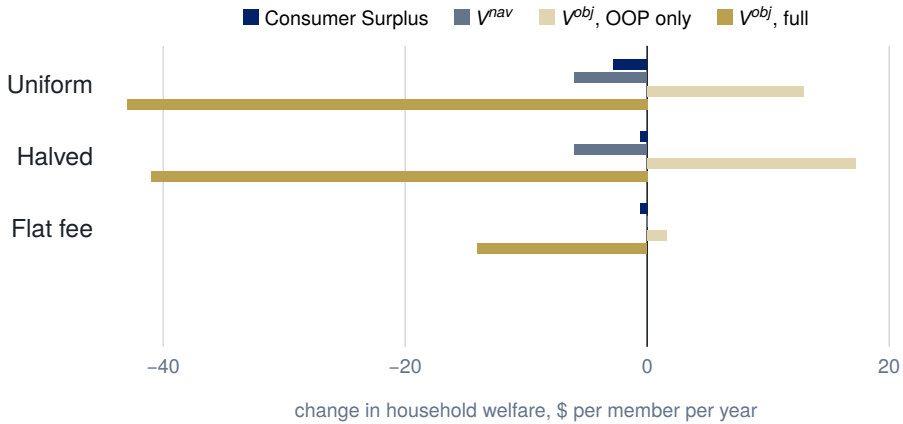
Counterfactual 2: Changes to Commission Schedule



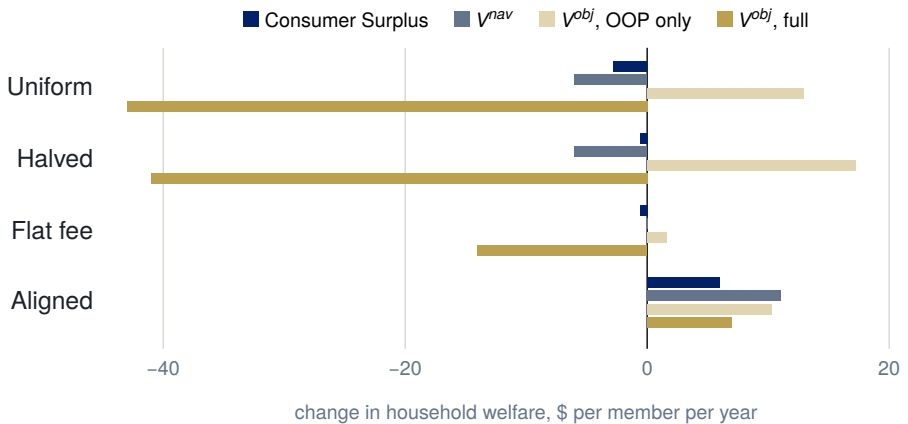
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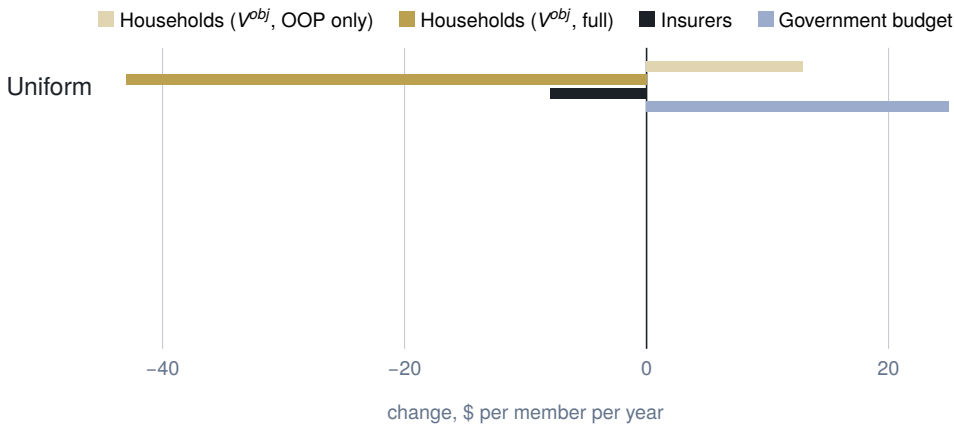
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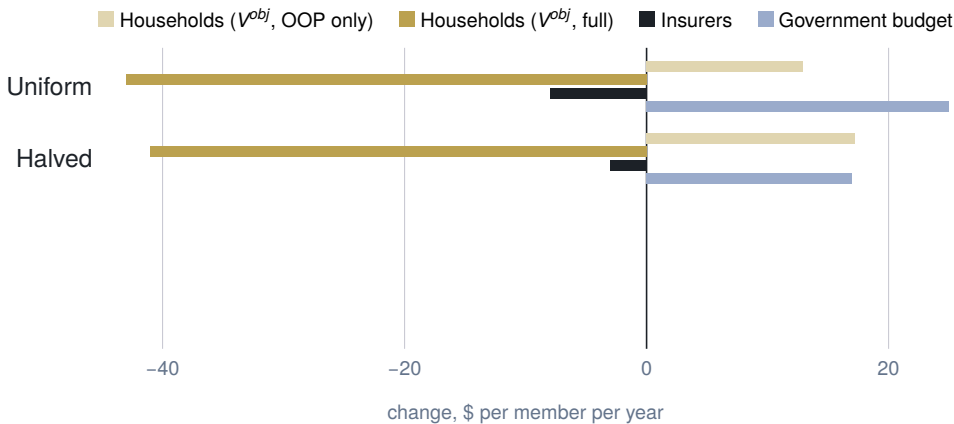
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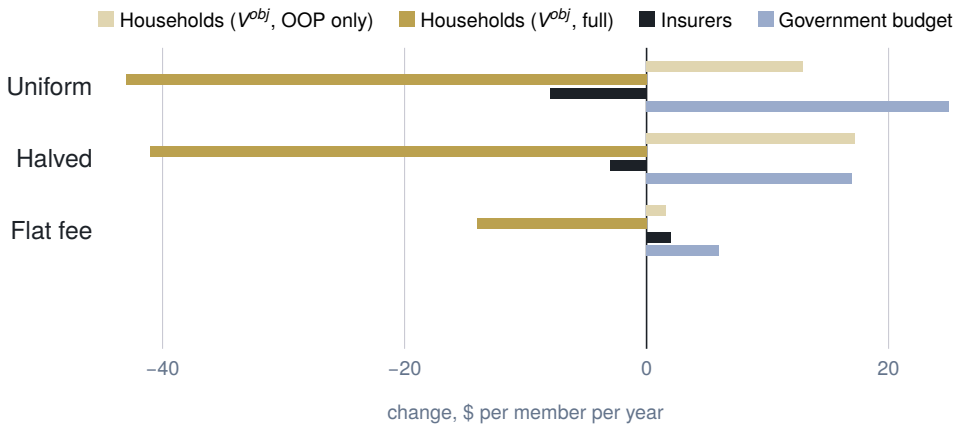
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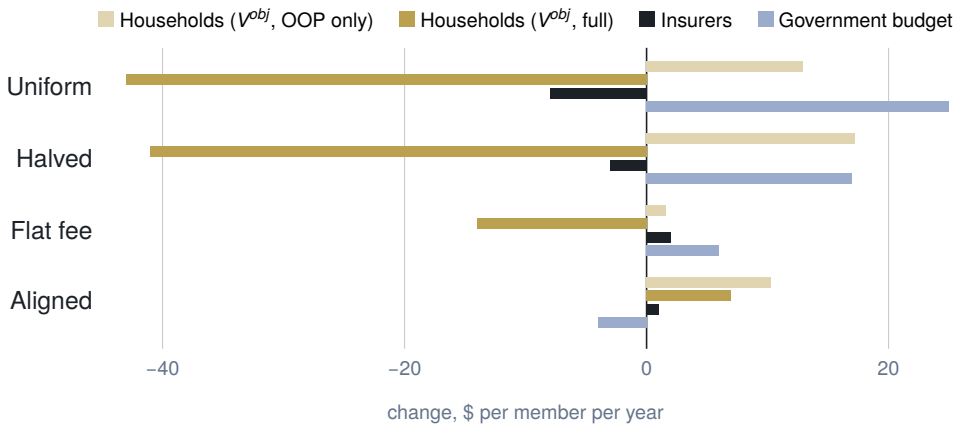
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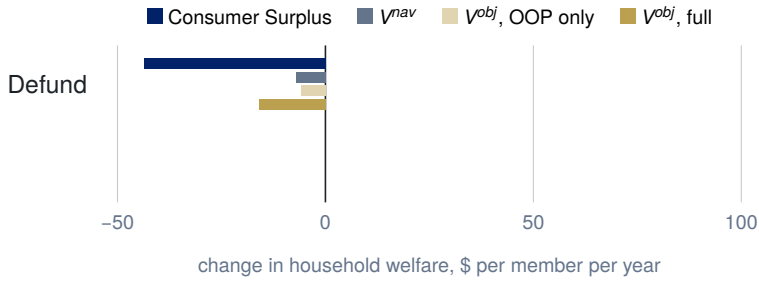
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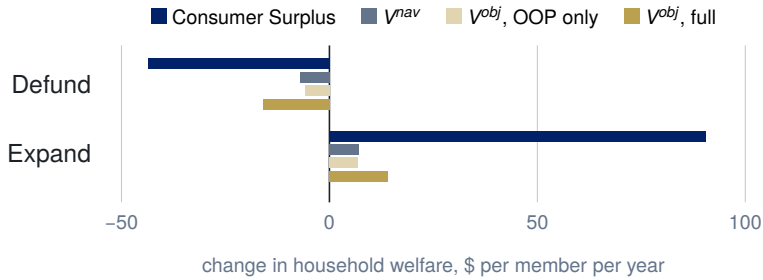
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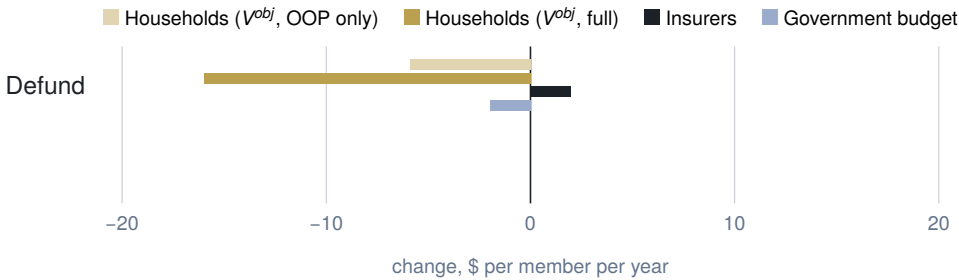
Counterfactual 3: Defunding or Expanding Navigators



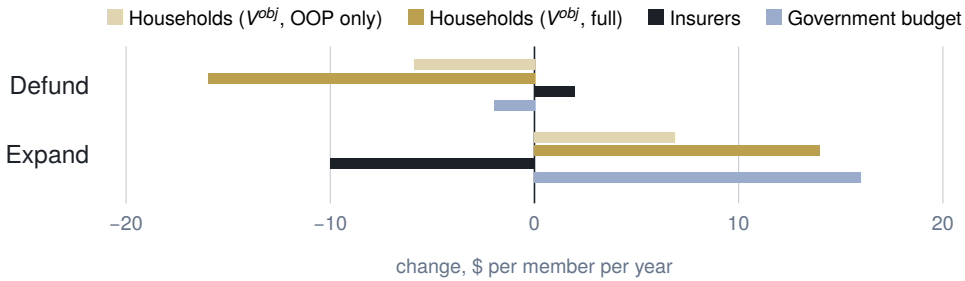
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Summary of Counterfactuals

	Households (V^{obj})	Insurers	Government budget
Ban commissions ($\tau = 0$)	–\$412	–\$51	+\$100
Ban commissions ($\tau = 100\%$)	+\$34	–\$24	+\$40

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Ban commissions ($\tau = 100\%$)	+$34	–$24	+$40
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Halved commissions	–$41	–$3	+$17
Aligned commissions	+$7	+$1	–$4
Flat fee	–$14	+$2	+$6

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Defund navigators	–$16	+$2	–$2
Expand navigators	+$14	–$10	+$16

Conclusion

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- Changes to the commission schedule change the uninsured share by at most 0.7pp. Aligning commissions with household value is the only change that raises all household welfare measures.
- Expanding navigator assistance raises household welfare and lowers subsidy spending, not counting the cost of the additional navigators.

Final Takeaways

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2. Changes to how agents are paid have small effects on coverage and household welfare compared with changes in access to assistance.

Thank You

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Robustness: excluding plan-based enrollers

Plan-based enrollers work for a single insurer, so their clients can only buy that insurer's plans. They account for 1.3% of household-years.

	All households	Excluding plan-based enrollers
Commission (agent clients)	0.027	0.030
Commission^2/100 (agent clients)	−0.066	−0.070
Premium (per $100)	−0.418	−0.420
Actuarial value	4.43	4.44
Nesting parameter λ	0.377	0.378

back to estimation

Gatekeeper simulation

- Agents may steer by showing clients only some plans, rather than by changing how clients value plans
- We simulate choices from a gatekeeper who shows a menu tilted toward high-commission plans, with no commission in utility, and fit our model to those choices
- Our model reproduces the gatekeeper's plan shares, with a largest gap of 1.1pp (1.7pp when the menu also depends on unobserved household or agent types) and a correlation above 0.997
- With commission in utility, the model predicts how choices change when commissions change. A gatekeeper model would also need to model which insurers agents contract with as commissions change

  Li et al. (WP) model how brokers contract with insurers based on commissions, in New York's small-group market

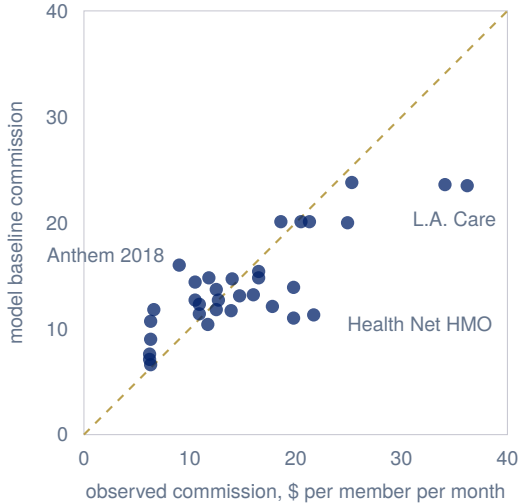
back to estimation

Three-level nested logit

- Households choose a channel on the value of each channel's menu (λ_C of 0.21 and 0.29)
- Removing agents raises the uninsured share 2.4–2.9pp, matching the scenario in which roughly half of agent clients switch to navigators
- But zero commissions would cut agent use 13–16 points, which the data across and within regions do not show

back to estimation

Model fit: commissions



Model and observed commissions have a correlation of 0.82 across 33 carrier-years, with a mean absolute difference of $3.32 on a mean commission of $15.

Model commissions are low for Health Net HMO (2015–17) and L.A. Care (2017–18), and high for Anthem and Blue Shield PPO in 2018.

back to commission FOC

Model fit: marginal cost

Marginal cost implied by inverting the pricing conditions has a correlation of 0.48 across plan-years with marginal cost from the claims equation, net of risk adjustment and reinsurance.

back to pricing FOC

Premiums and subsidies

$$p_{ijt} = \max\left\{ \overbrace{\sigma_{it}\, p_{jmt}}^{\text{full premium}} - \overbrace{\max\{\sigma_{it}\, p_{bmt} - \zeta_{it},\, 0\}}^{\text{subsidy}},\ \overbrace{0}^{\text{floor}} \right\}$$

The net premium household *i* pays for plan *j* in rating region *m*, year *t*

back to setting

Premiums and subsidies

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- p_{jmt} is the plan's base premium in the household's rating region
- σ_{it} is the household's age rating factor (a 64-year-old pays up to three times what a 21-year-old pays)

back to setting

Premiums and subsidies

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- Available for incomes of 100–400% of FPL
- p_{bmt} is the base premium of the benchmark plan, the second-cheapest silver plan in the region
- ζ_{it} is the household's contribution cap, from 2% of income at 100% of FPL to 9.5% at 400% (2014)
- The same dollar subsidy applies to any plan

back to setting

Premiums and subsidies

$$p_{ijt} = \max\left\{ \overbrace{\sigma_{it}\, p_{jmt}}^{\text{full premium}} - \overbrace{\max\{\sigma_{it}\, p_{bmt} - \zeta_{it},\, 0\}}^{\text{subsidy}},\ \overbrace{0}^{\text{floor}} \right\}$$

- When the subsidy exceeds a plan's full premium, the household pays nothing
- Some bronze plans are free to low-income households

back to setting

Premiums and subsidies

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- Raising the benchmark premium raises every household's subsidy
- The benchmark's own net price is unchanged for subsidized households
- Every rival plan's net price falls

back to setting

Premiums and subsidies: an example

A single 60-year-old at 200% of FPL in Los Angeles, 2016

- Age rating factor $\sigma = 2.714/1.278 = 2.12$ relative to age 40

back to setting

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back to setting

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back to setting

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	Base (age 40)	Full (age 60)	Net premium
Molina bronze	$191	$405	**$0**
Health Net silver (benchmark)	$255	$541	$126
Kaiser silver	$312	$663	$247
Molina gold	$265	$562	$146
Blue Shield gold	$402	$853	$438

back to setting

Identification of key parameters

- **Premium:** variation across plans, the mandate penalty phasing in (2014–2016) and ending (2019), and kinks in the subsidy formula that make some plans free
- **Assistance terms:** assisted and unassisted households selecting different plans from the same menu within a region and year
- **Commission terms:** commissions differ across insurers at a given actuarial value and change over time within some insurers
- **Insurer effects:** brand indicators absorb networks and other unobserved features, so commission effects come from within-insurer variation

back to estimation

Open questions

Do households choose their assistance channel?

- **Data limitation:** we only observe the channel for enrolled households, so we cannot estimate a model of channel choice for uninsured households
- A three-level nest allowing for channel choice assumes that higher commissions also raise agent use
- Across regions, agent use is not strongly correlated with commissions or with number of plans available, so it's not clear that households are making much of a "choice" here

back to model

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How do insurers set commissions?

- We assume each carrier sets commissions to maximize profit, weighing the profit from steered enrollment against the net cost of the commission
- The commission this implies is very sensitive to estimated margins, since a $10 change in margins changes it by about as much as the average commission ($13)
- So the condition fits observed commissions only with carrier constants δ_f, which absorb much of the variation across carriers
- ...but most commission variation is across carriers and constant over time anyway

back to model

Summary of Counterfactuals

	Households	Government	
Defund navigators	−$16	+$2	−$2
Expand navigators	+$14	−$10	+$16

Insurers

- ↑ lower commission payments
- ↓ less steering toward higher-commission plans
- ↓ fewer enrollees
- ↓ lower risk adjustment as the remaining pool gets riskier
- ↓ lower premiums

Summary of Counterfactuals

	Households (V^{obj})	Insurers	Government budget
Defund navigators	−$16	+$2	−$2
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Government budget

- ↑ fewer enrollees receiving tax credits
- ↑ lower benchmark premiums, so smaller tax credits
- ↑ more penalty revenue from the uninsured
- ↓ more uncompensated care for the uninsured